

Splunk SOC Home Lab – Windows Security Monitoring and Detection

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Tools Used:

- Splunk Enterprise
- Windows 10
- Sysmon
- Universal Forwarder
- Parrot Security OS
- FreeRDP

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Executive Summary

This project demonstrates the implementation of a Security Operations Center (SOC) home laboratory using Splunk Enterprise. Windows Security logs and Sysmon events were collected and analyzed to detect common security activities.

Several security scenarios were simulated, including user account creation, account deletion, privilege escalation, failed login attempts, successful authentication events, Powershell Execution and Remote Desktop Protocol (RDP) access. Splunk searches were used to investigate and validate each activity.

The project demonstrates practical SOC analyst skills involving log monitoring, event investigation, threat detection, and security reporting.

1. Lab Environment

Description –

A Windows 10 virtual machine was configured as the monitored endpoint. Splunk Enterprise was used as the Security Information and Event Management (SIEM) platform. Windows Event Logs and Sysmon logs were forwarded to Splunk for centralized monitoring.

```
Microsoft Windows [Version 10.0.19045.0450]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>hostname
DESKTOP-PF3D72U

C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2401:4000:8f0e:c205:196d:1d4a:a31f:aa59
Temporary IPv6 Address. . . . . : 2401:4000:8f0e:c205:1d06:c27d:9c7b:falc
Link-local IPv6 Address . . . . . : fe80::1798:8943:9ed1:b7ec%5
IPv4 Address. . . . . : 192.168.1.2
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::260a:8aff:feb4:0011%5
                               192.168.1.1

C:\Windows\system32>
```

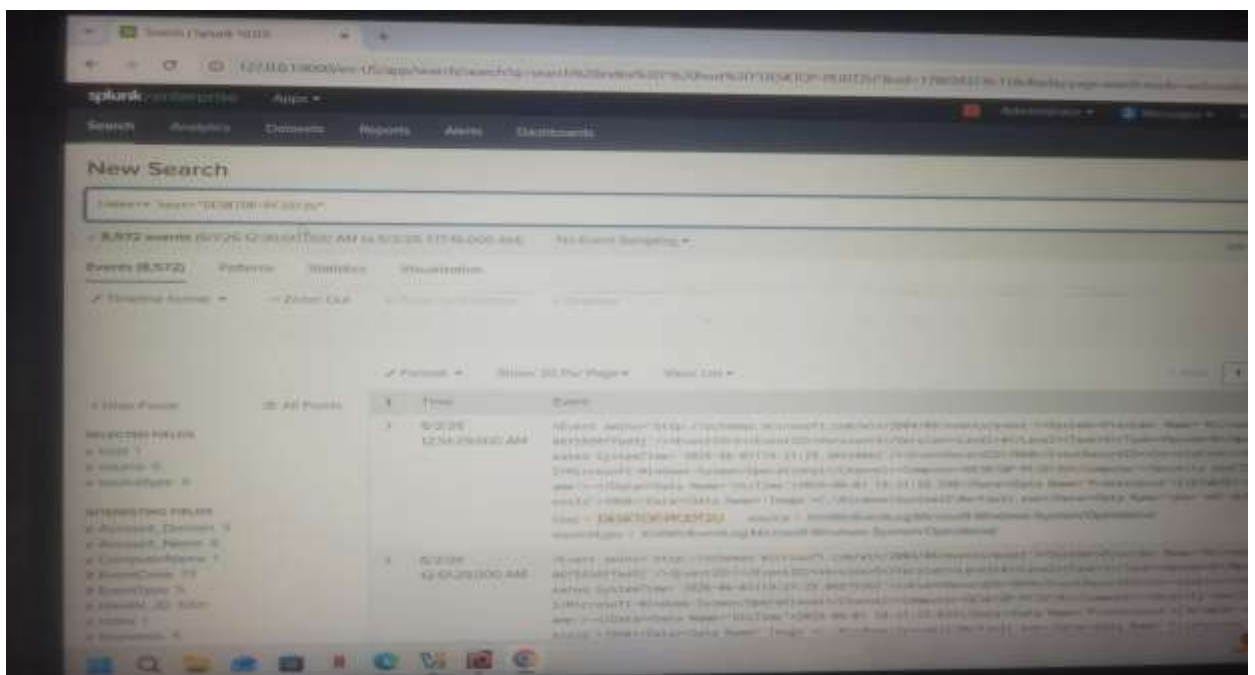
Observation –

The endpoint hostname and IP address were verified before beginning security monitoring activities.

2. Log Collection Validation

Description –

Before conducting security tests, log ingestion was verified to ensure that Windows events were successfully reaching the Splunk server.



➤ **Splunk Query –**

index=* host="DESKTOP-PFJDT2U

Result –

More than 8,000 events were successfully indexed, confirming proper log collection.

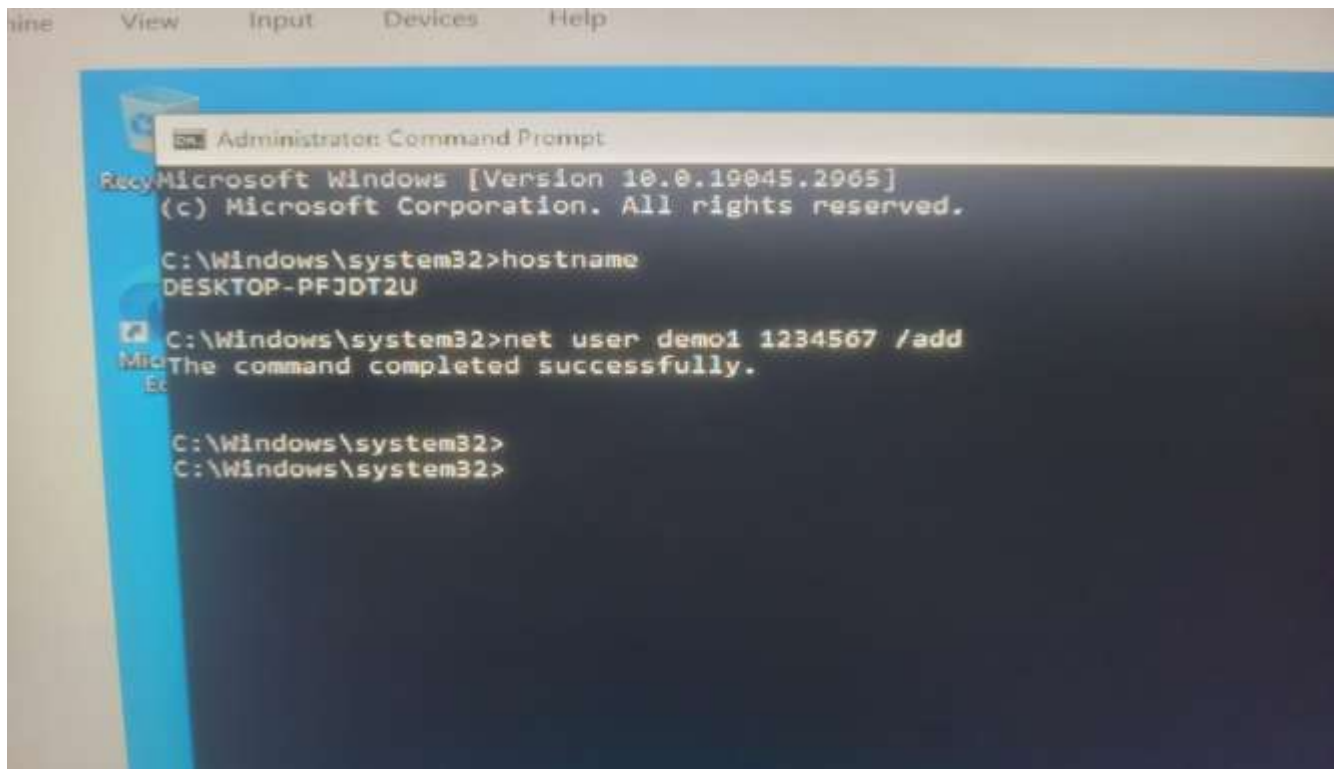
3. Scenario 1 – User Account Creation Detection (Event ID 4720)

Objective -

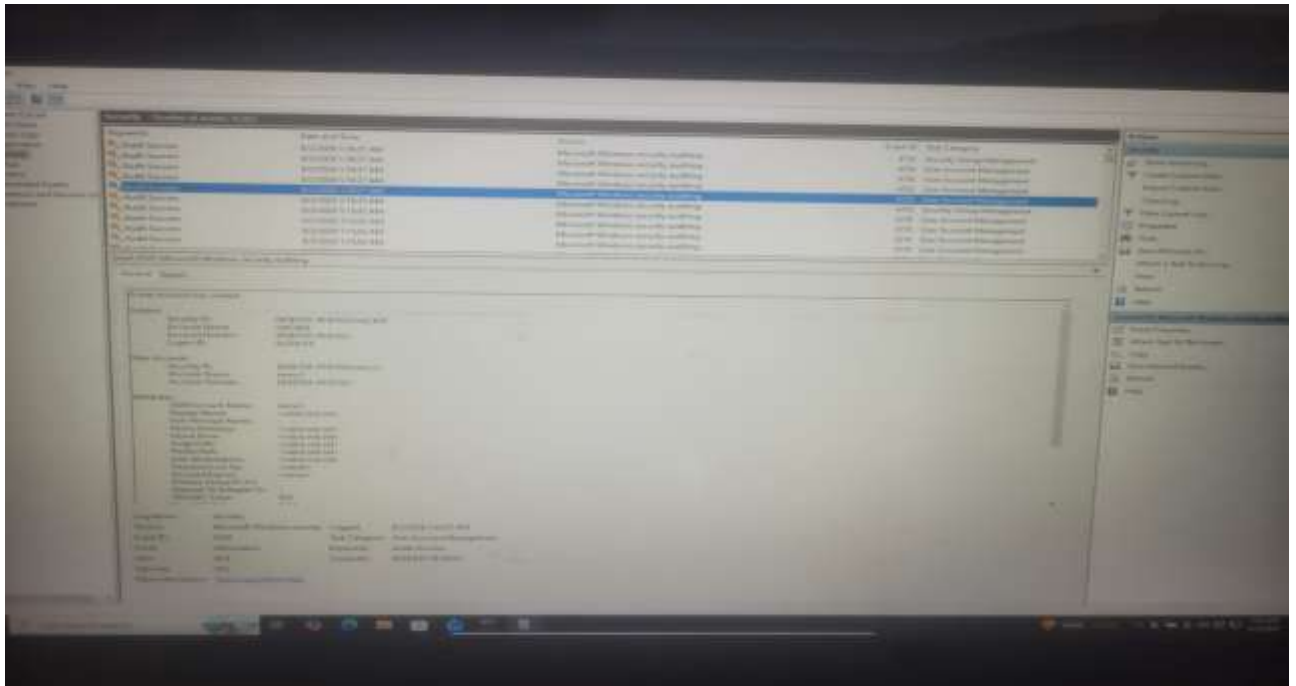
Detect creation of a new Windows user account.

Activity Performed –

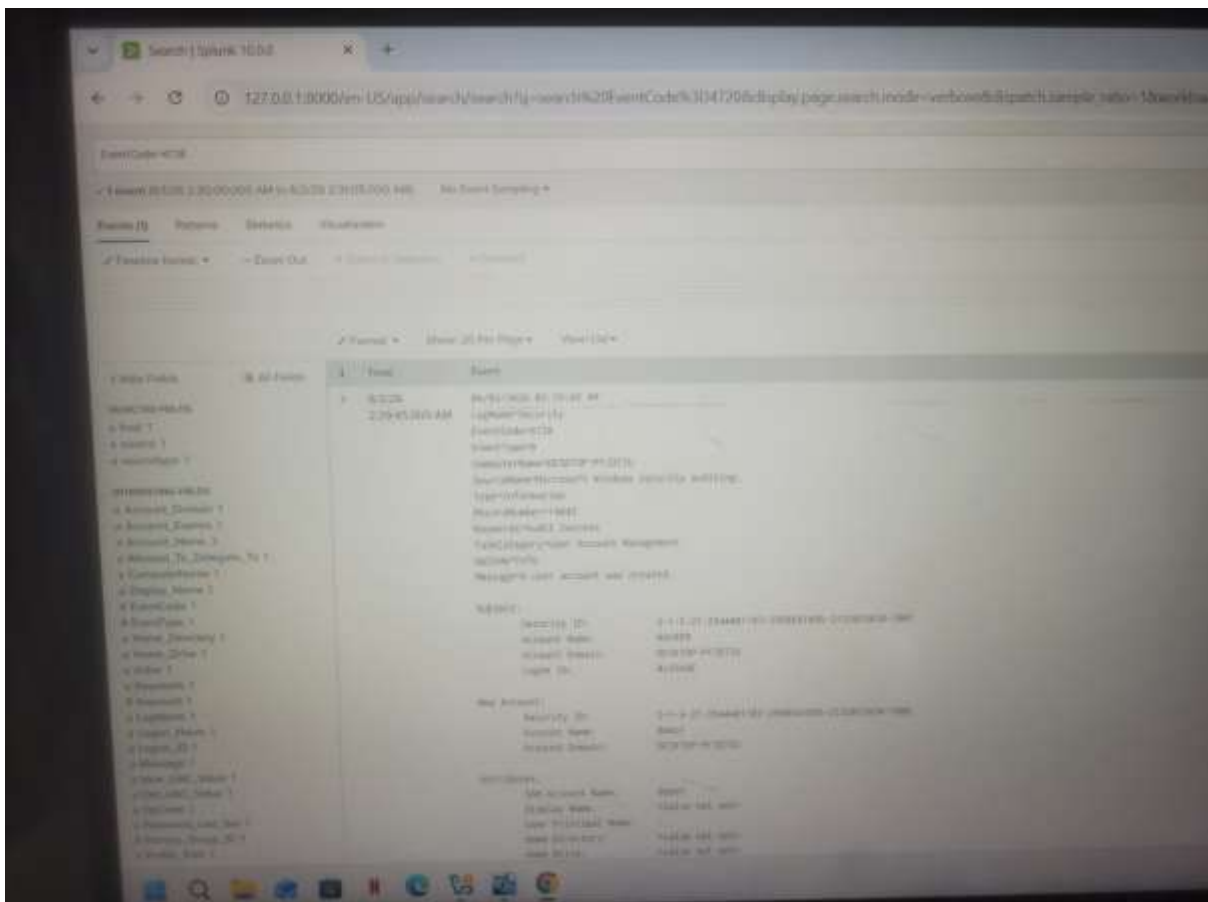
A new local user account named "demo1" was created using the Windows command line.



Event Verification –



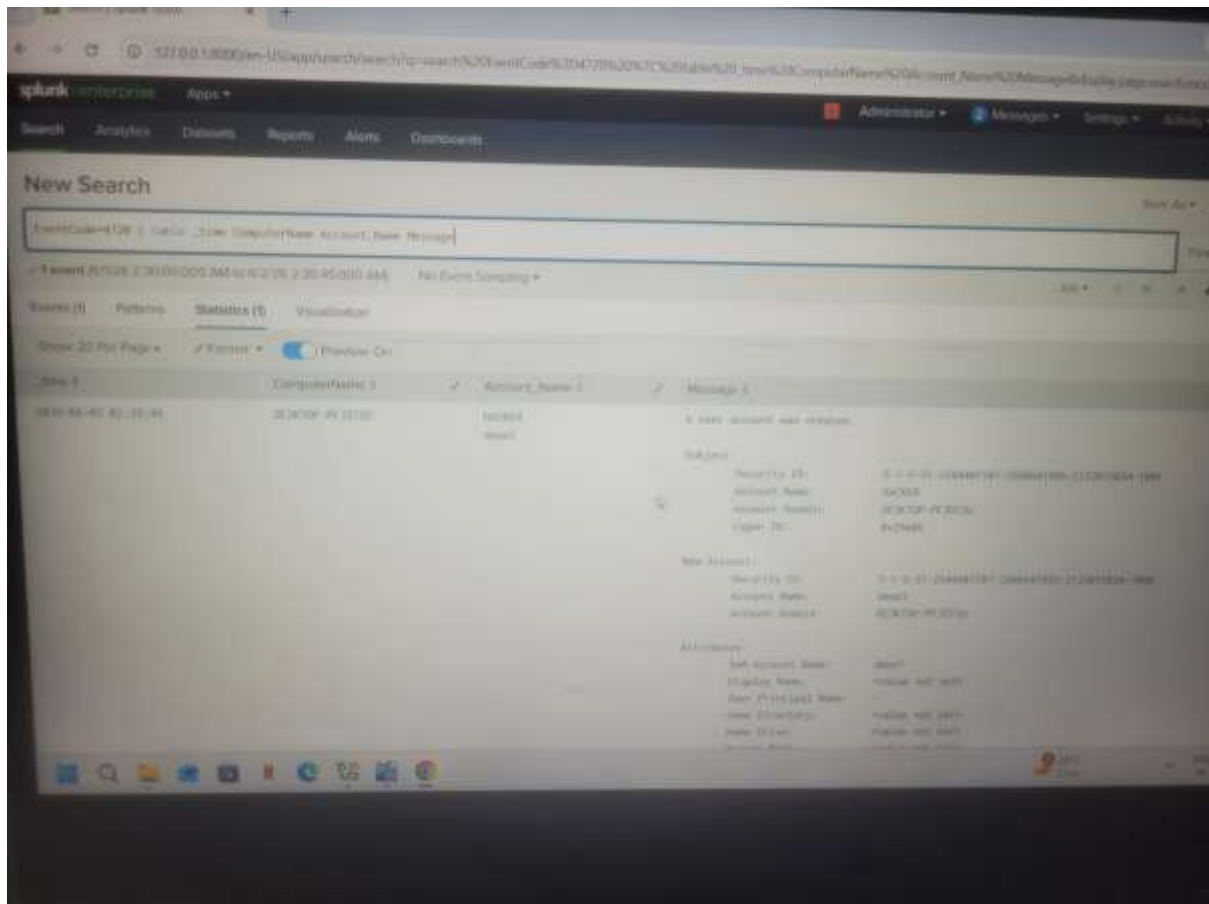
➤ Splunk Search – EventCode=4720



Analysis –

Event ID 4720 indicates that a new user account was created. Such events are important because attackers often create new accounts to establish persistence.

Investigation View –

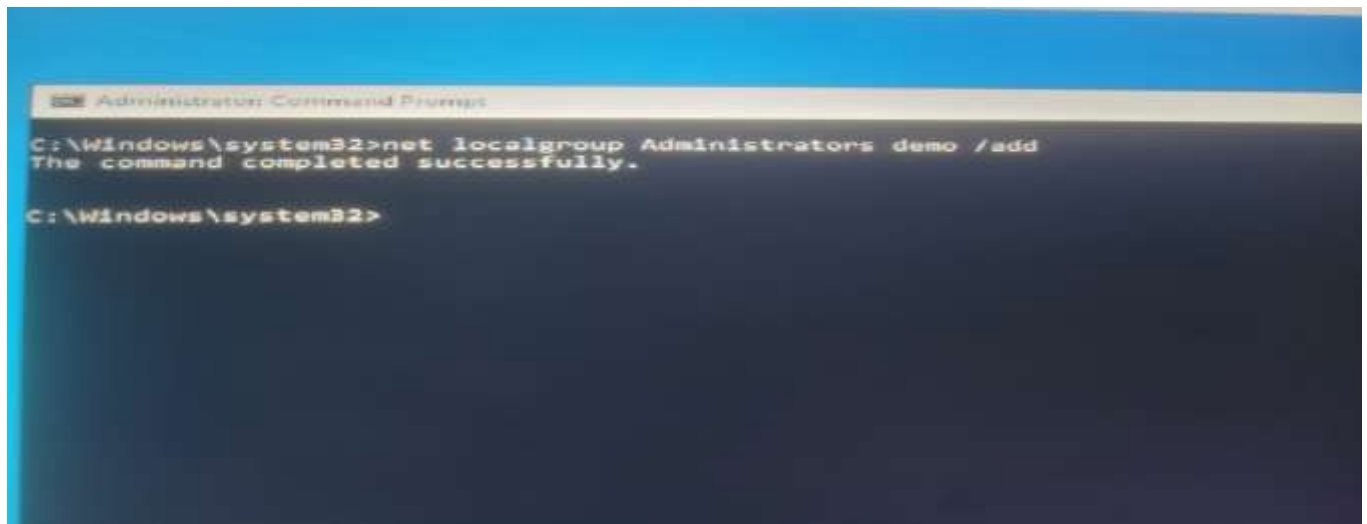


4. Scenario 2 – Privilege Escalation Detection (Event ID 4732)

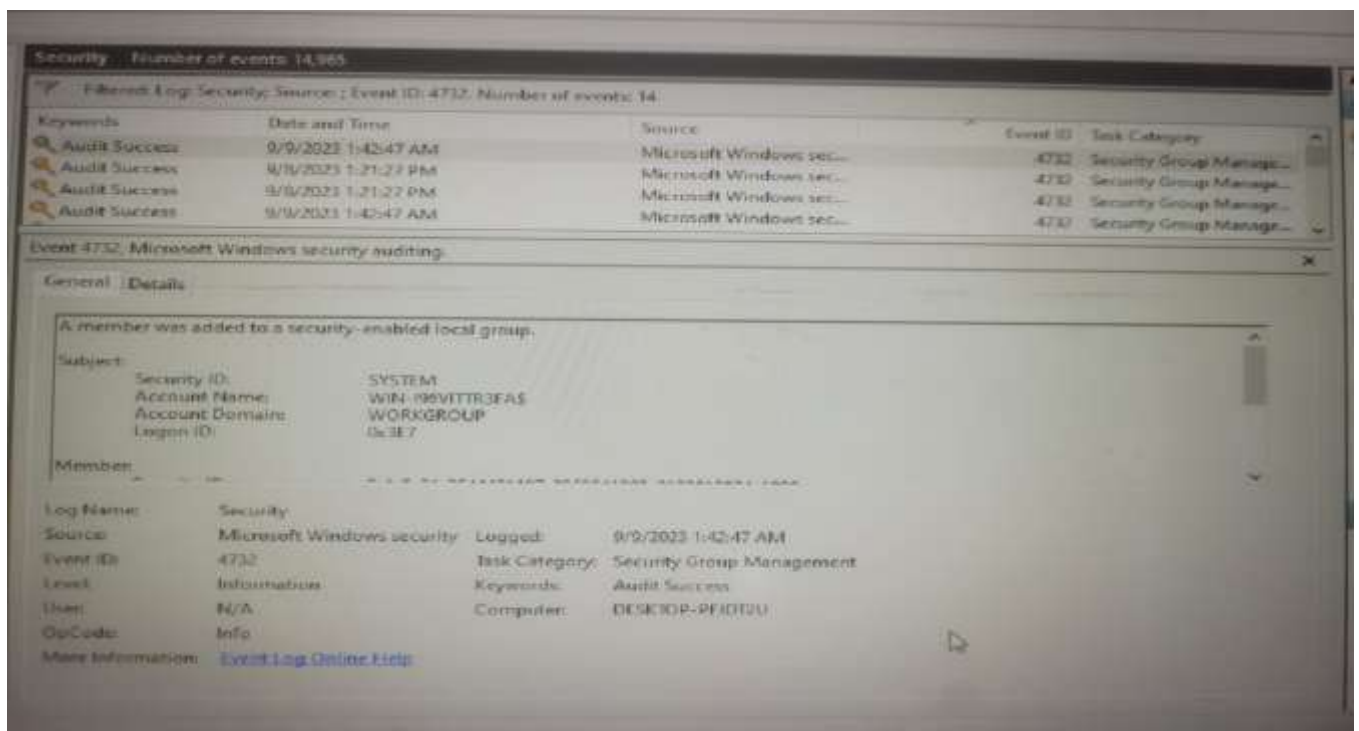
Objective -

Detect addition of a user to the local Administrators group.

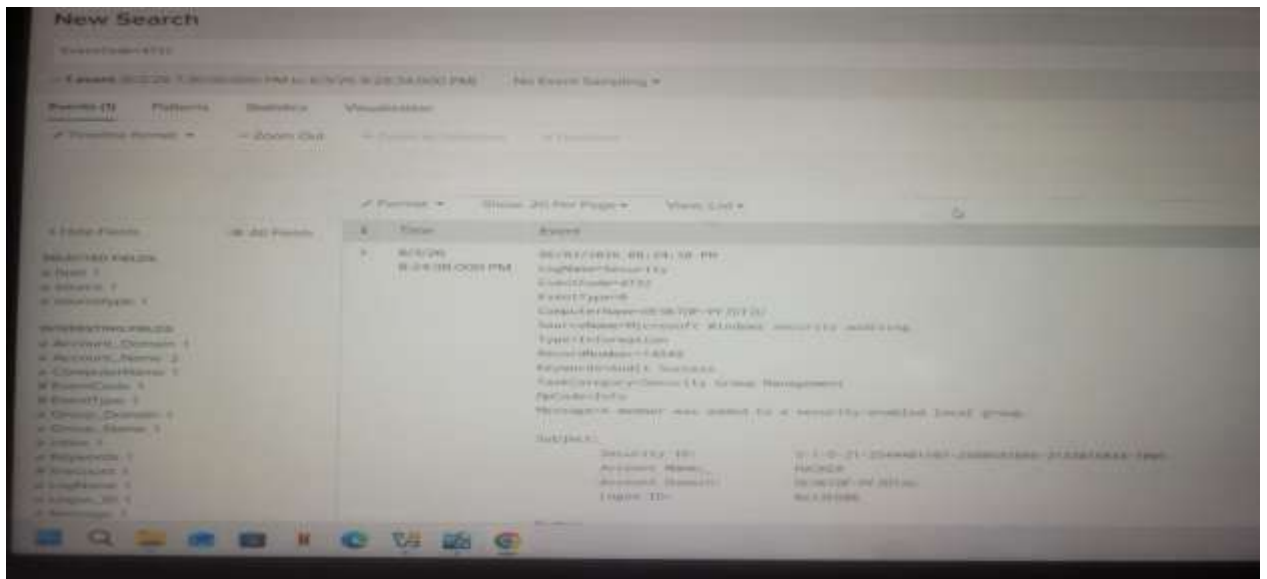
Activity Performed –



Event Verification –



➤ Splunk Search – EventCode=4732



Analysis –

Event ID 4732 indicates that a member was added to a privileged local group. Monitoring this event is critical because privilege escalation is commonly used by attackers after gaining access.

5. Scenario 3 – Failed Login Detection (Event ID 4625)

Objective –

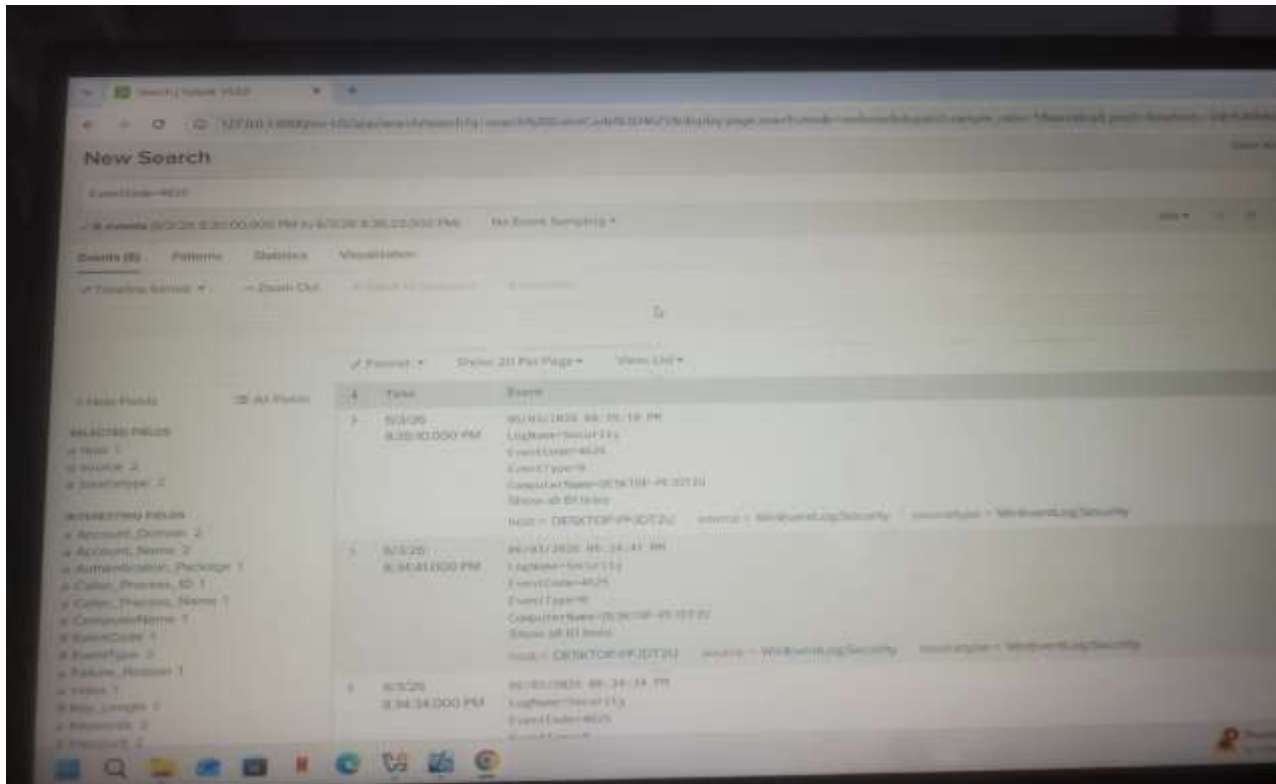
Detect unauthorized login attempts.

Activity Performed –

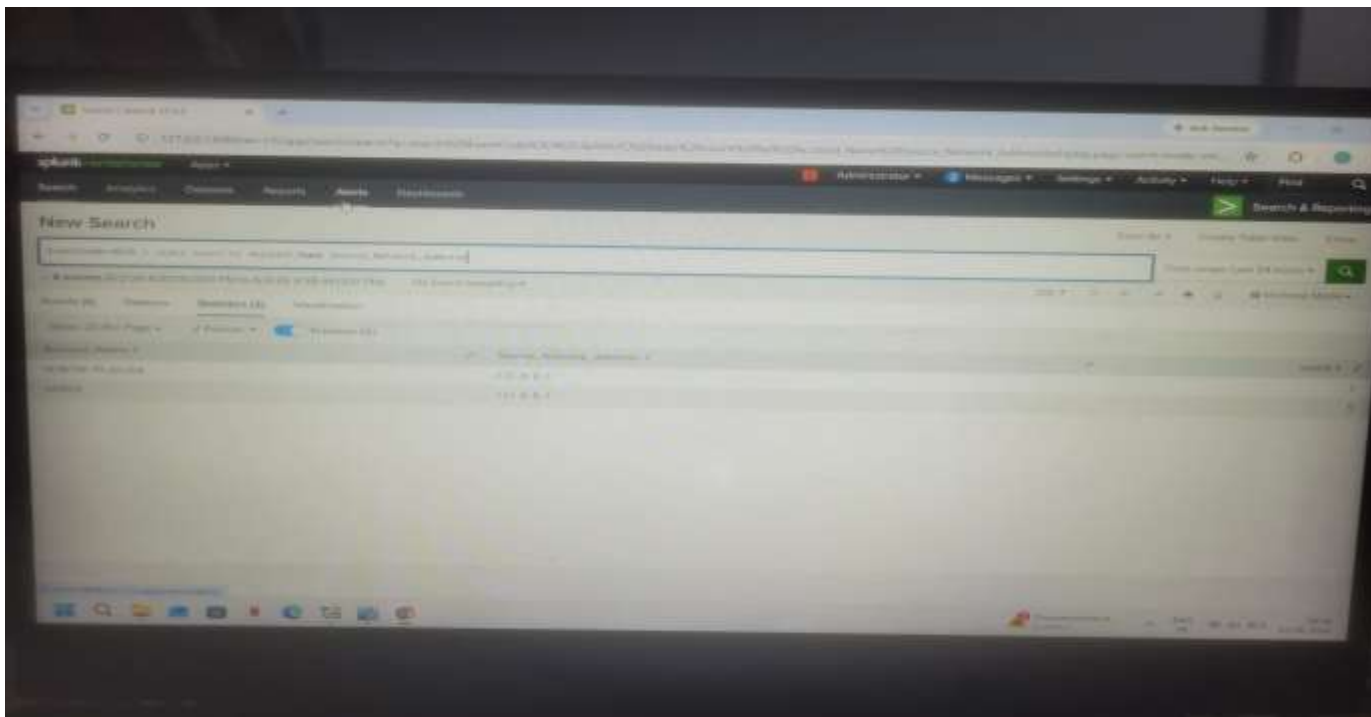
Multiple login attempts were performed using incorrect credentials.



➤ Splunk Search – EventCode=4625



➤ Statistical Analysis – EventCode=4625 | stats count by Account_Name Source_Network_Address



Analysis –

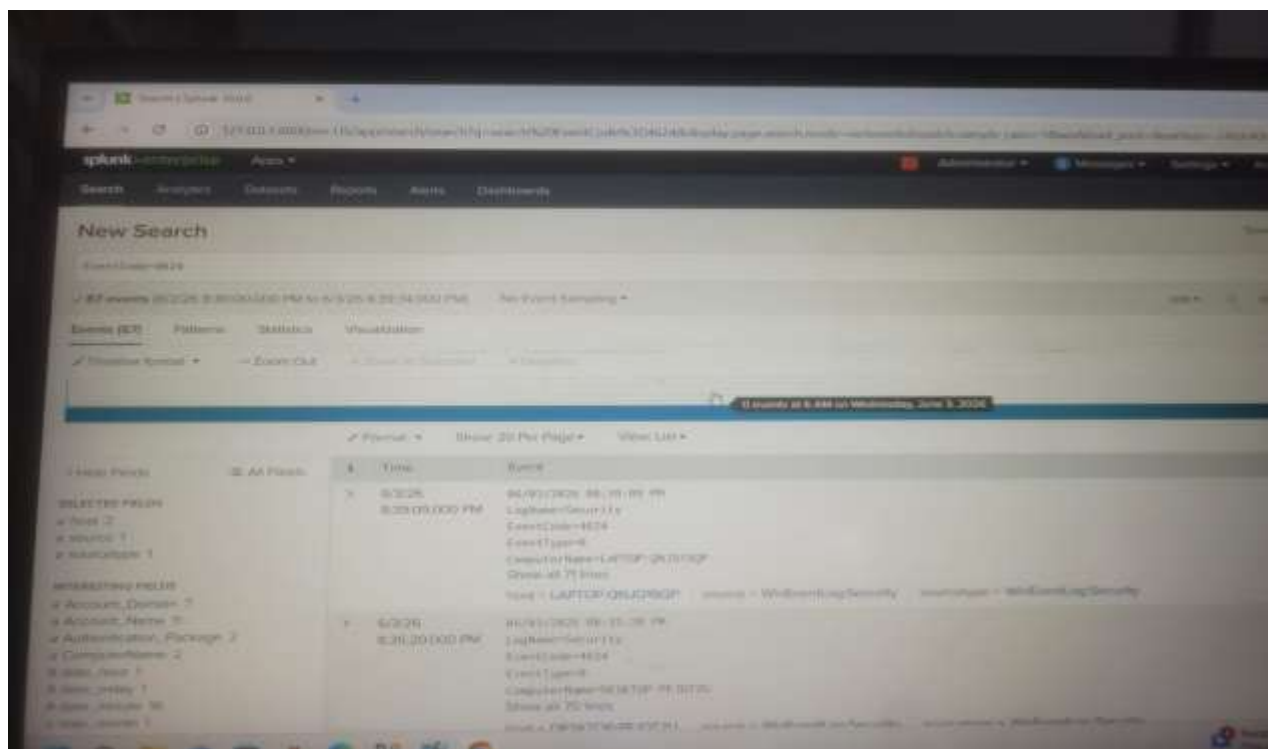
Multiple failed authentication attempts may indicate password guessing or brute-force activity. Splunk successfully detected and aggregated these events.

6. Scenario 4 – Successful Login Detection (Event ID 4624)

Objective –

Monitor successful user authentication events.

➤ Splunk Search – EventCode=4624

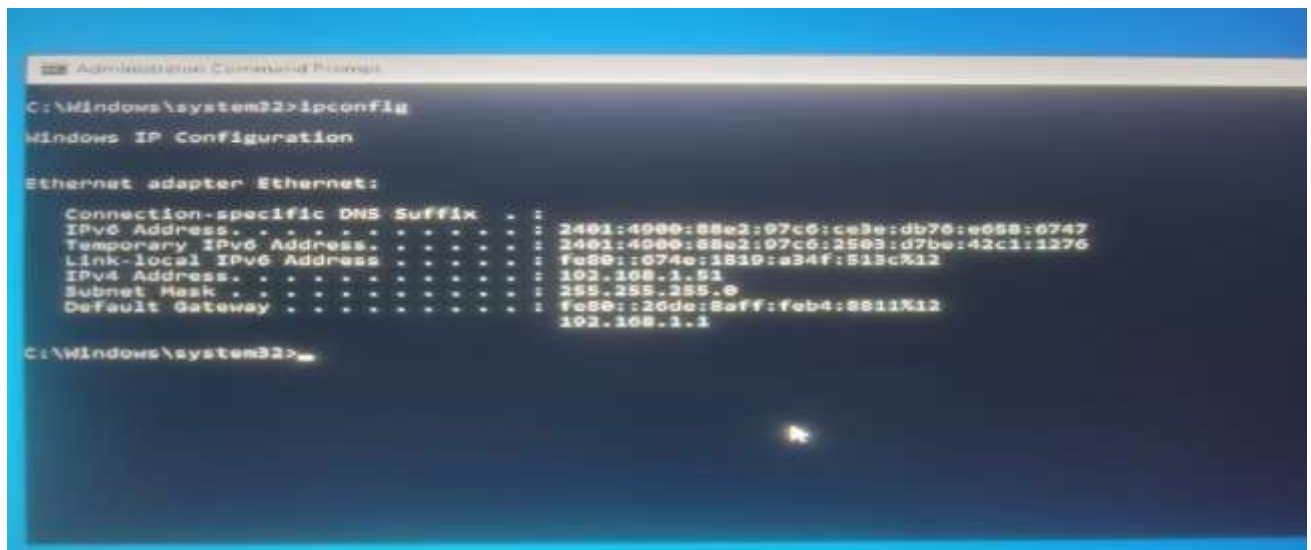


Analysis –

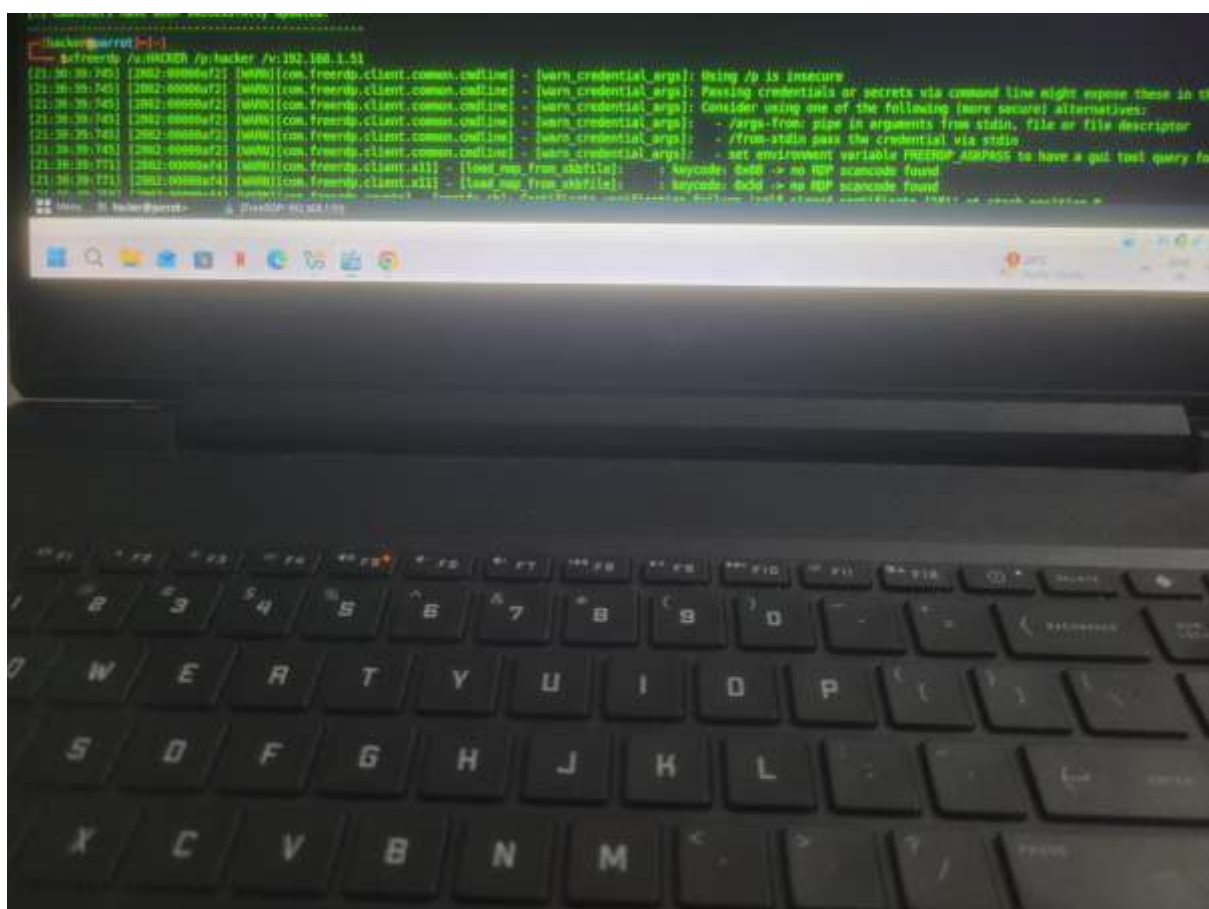
Event ID 4624 confirms successful authentication and can be used to establish user activity timelines.

7. Scenario 5 – RDP Monitoring

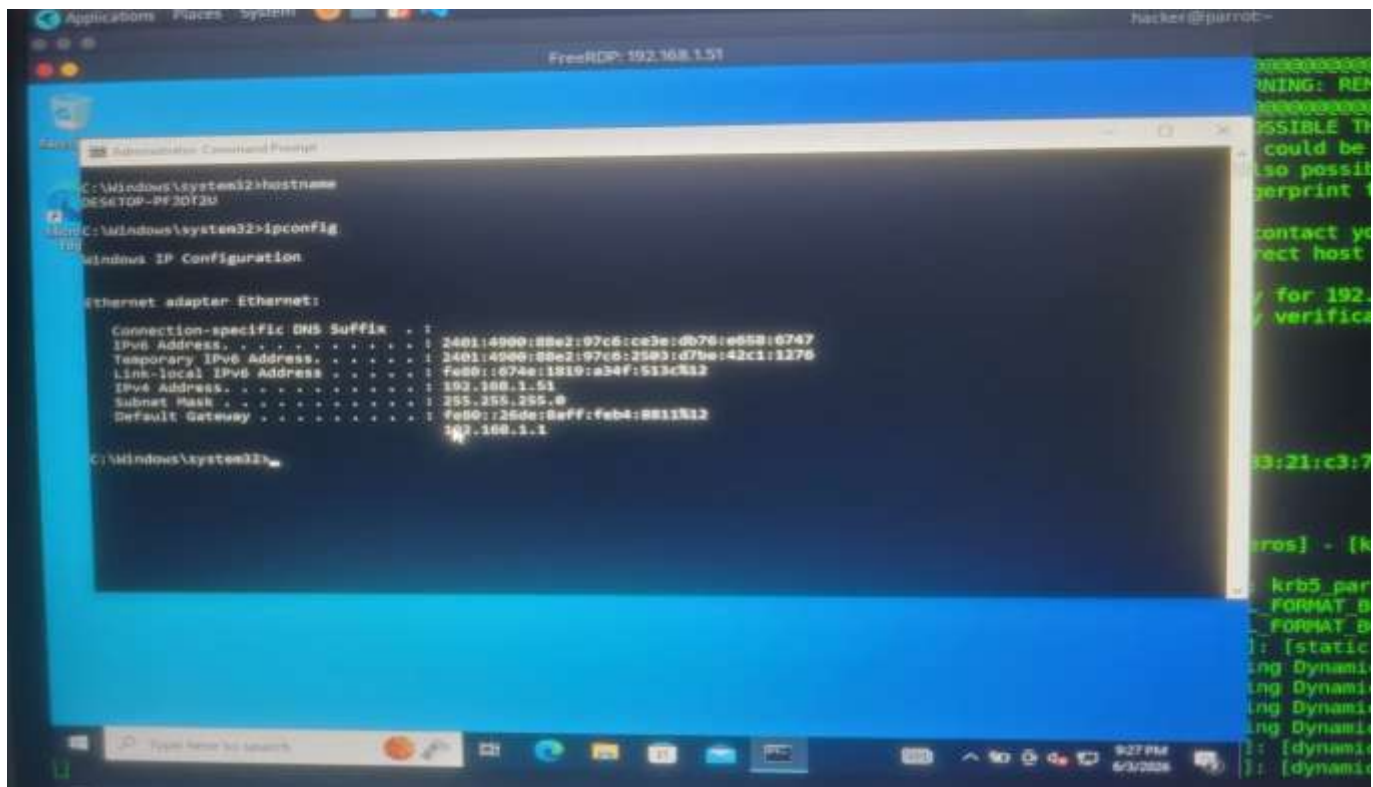
Host Identification –



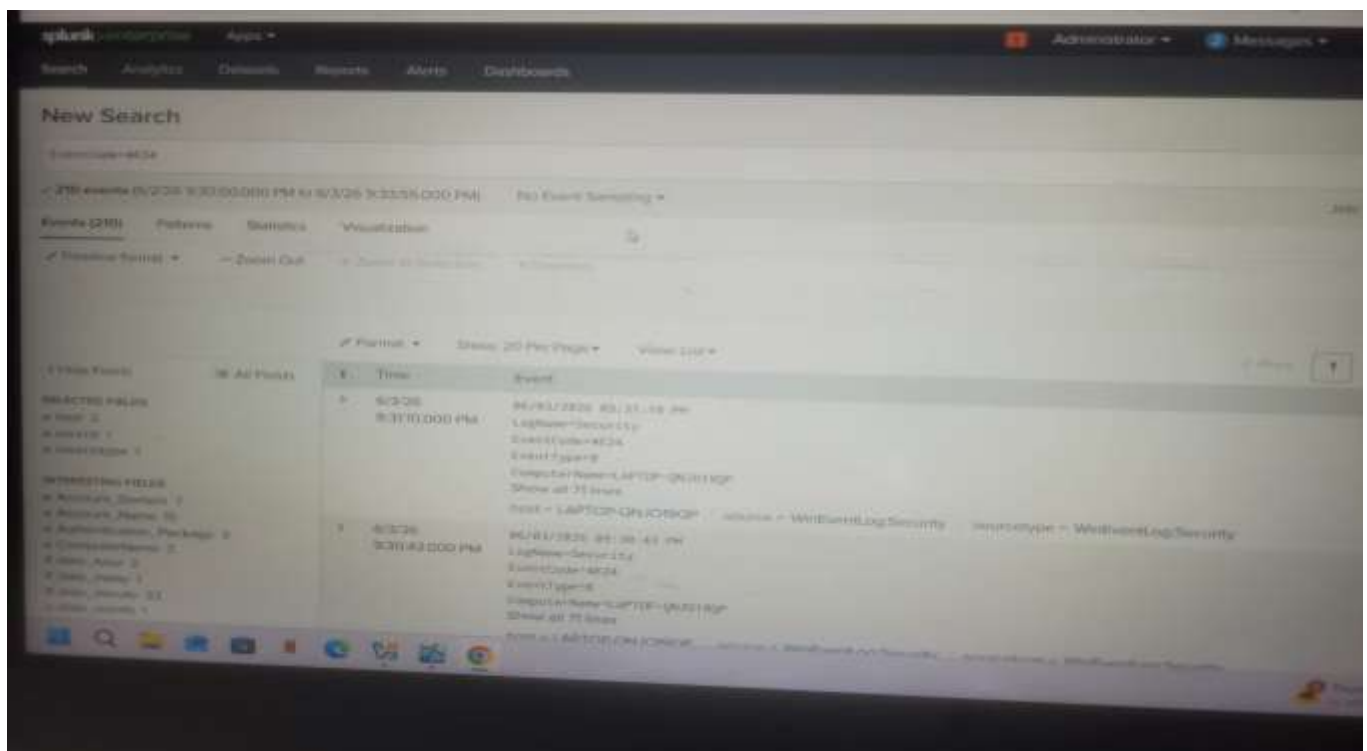
RDP Connection Attempt –



Successful Connection –



Splunk Validation –



Analysis –

Remote Desktop access was successfully established and corresponding authentication events were captured in Splunk. RDP

monitoring is important because remote access services are frequently targeted by attackers.

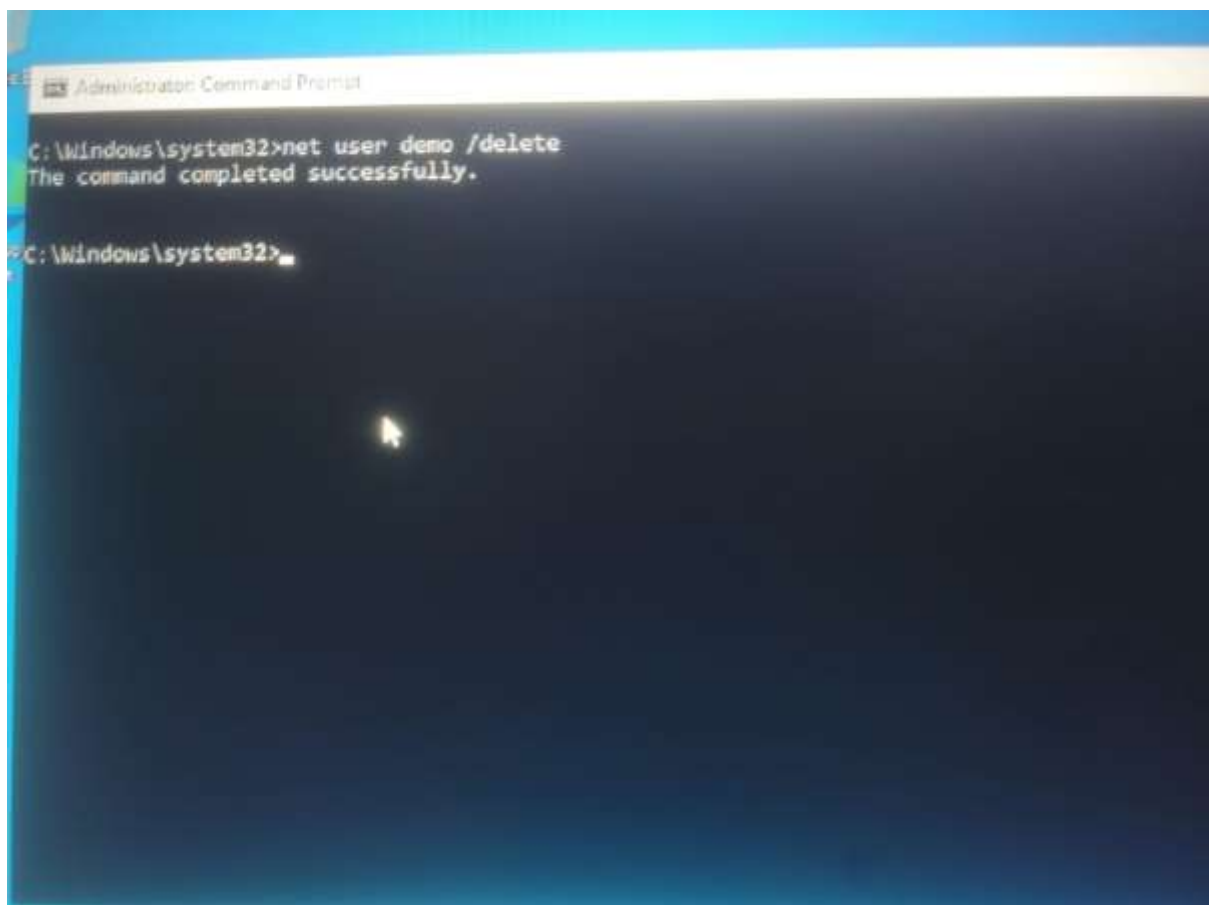
8. Scenario 6 – User Account Deletion Detection (Event ID 4726)

Objective –

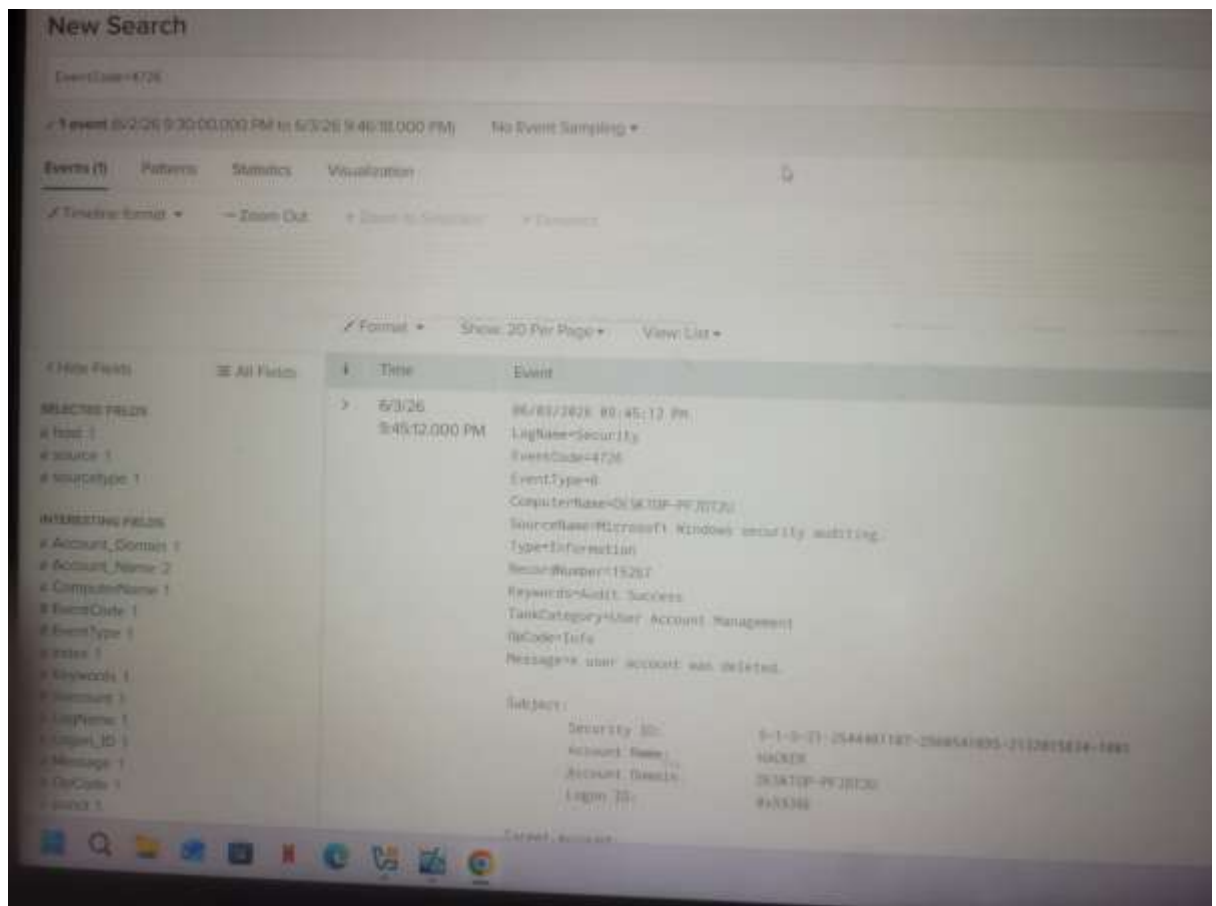
Detect deletion of a Windows user account using Windows Security Event Logs.

Activity Performed –

A previously created local user account was deleted from the Windows system.



- **Splunk search –**
EventCode=4726



Analysis -

Event ID 4726 indicates that a user account was deleted from the system.

Monitoring account deletion events is important because attackers may remove accounts to hide their activities or administrators may remove compromised accounts during incident response activities.

9. Scenario 7 – PowerShell Activity Monitoring

Objective –

Monitor PowerShell activity and detect command execution through Windows PowerShell Operational logs.

Activity Performed –

Several PowerShell commands were executed to generate PowerShell Operational log events.

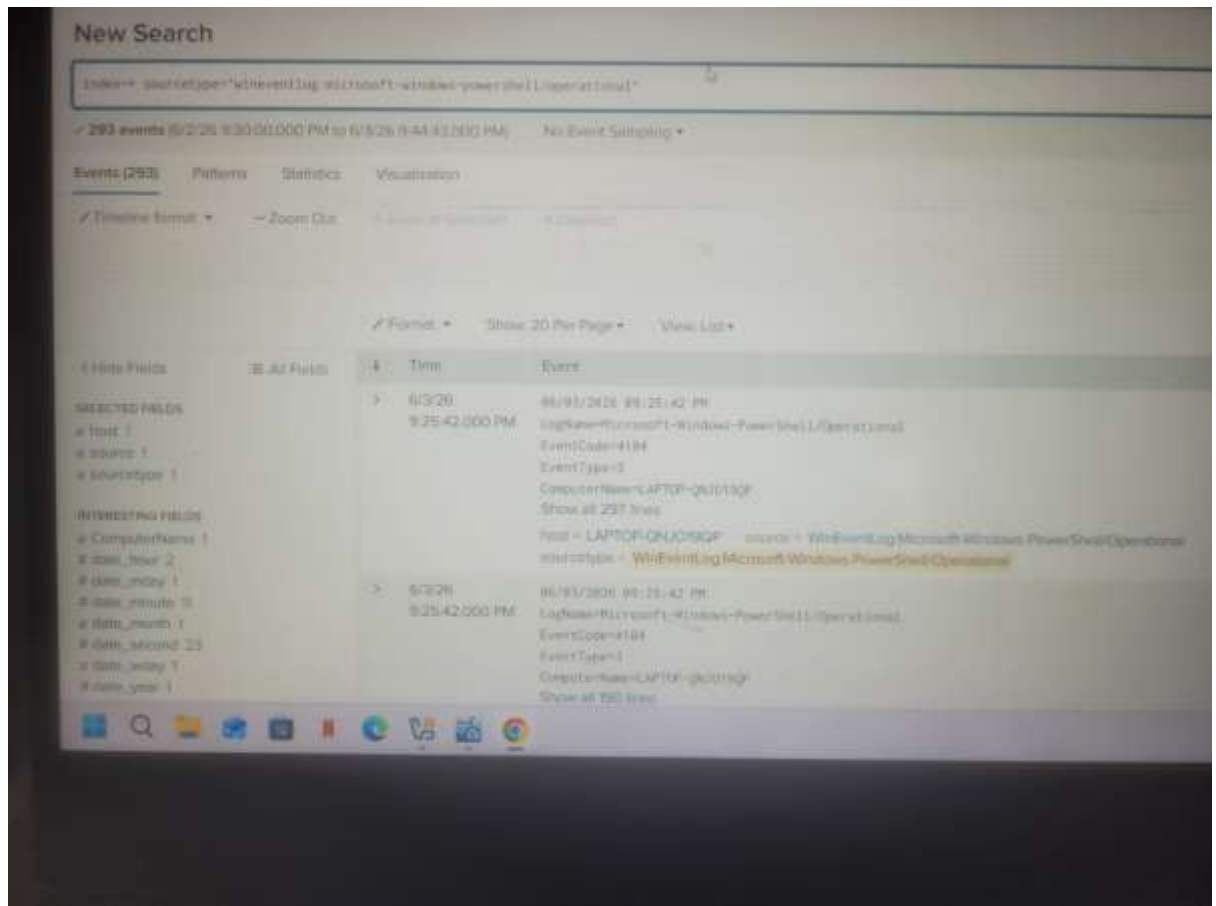
```

PS C:\Windows\system32> Get-Process

Handles NPM(K) PM(K) WS(K) CPU(s) Id SI ProcessName
-----
359 21 5472 22080 1.72 8048 1 ApplicationFrameHost
176 11 8496 12184 0.28 8712 0 audiodg
70 5 2472 1920 0.44 7196 1 cmd
157 10 8532 1444 1.14 3188 0 conhost
262 14 4164 18584 0.25 5940 1 conhost
260 14 7392 13888 4.84 7960 1 conhost
454 15 1760 2072 3.67 496 0 csrss
457 19 1956 3012 1,135.00 580 1 csrss
438 17 4240 21196 0.81 6352 1 ctfnun
190 16 3820 3092 0.56 2436 0 dlnhost
253 23 5692 8548 0.69 3876 1 dlhhost
985 47 41916 63920 15.99 428 1 dmw
2267 82 44612 81856 24.52 4332 1 explorer
36 6 1492 1572 0.45 864 1 fontdrvhost
36 5 1280 548 0.75 872 0 fontdrvhost
0 0 0 0 0 0 0 Idle
1340 25 6912 13304 8.78 732 0 lsass
1161 35 55848 30380 10.50 6744 1 RSASvc
0 0 488 77764 19.83 1624 0 Memory Compression
211 13 2896 1836 0.00 4028 0 MicrosoftEdgeUpdate
338 18 3888 2924 10.82 4172 0 MicrosoftEdgeUpdate
379 19 5820 5152 7.13 6716 0 MicrosoftEdgeUpdate
465 16 7468 8764 2.36 3448 0 MpDefenderCoreService
1210 42 48104 38448 12.72 4288 1 msedge
185 11 5640 4288 0.42 5604 1 msedge
329 17 12848 6724 1.45 3916 1 msedge
319 16 12664 9572 0.50 5628 1 msedge
173 10 2720 2964 0.51 3728 1 msedge
330 17 12820 10502 13.42 3468 1 msedgehnbview2
306 26 38388 12024 18.42 3844 1 msedgehnbview2
192 12 9780 3592 0.50 6432 1 msedgehnbview2
1105 40 48200 38420 11.67 6436 1 msedgehnbview2
370 21 17544 10220 0.89 6636 1 msedgehnbview2
171 10 2688 2912 0.27 6864 1 msedgehnbview2
671 229 244800 14240 50.30 2688 0 MsMpEng
955 53 30296 21584 17.92 4556 1 OneDrive
551 29 47872 13788 3.19 6792 1 OneDrive.Sync.Service
837 34 61848 78276 2.85 2994 1 powershell

```

- **Splunk Detection**
index=*sourcetype="wineventlog:microsoft-
windows-powershell/operational"



Findings –

Scenario	Event ID	Activity Detected	Detection Status
User Account Creation	4720	New user account created	Successfully Detected
User Account Deletion	4726	User account removed	Successfully Detected

Scenario	Event ID	Activity Detected	Detection Status
Privilege Escalation	4732	User added to Administrator's group	Successfully Detected
Failed Login	4625	Invalid login attempts	Successfully Detected
Successful Login	4624	User authenticated successfully	Successfully Detected
RDP Login	4624 (Logon Type 10)	Remote Desktop access	Successfully Detected
PowerShell Activity	4104 / Operational Logs	PowerShell command execution	Successfully Detected

Skills Demonstrated –

- SIEM Monitoring
- Log Collection and Forwarding
- Windows Event Log Analysis
- PowerShell Monitoring
- Event Investigation
- Detection Engineering

- Security Documentation
- Threat Detection

Conclusion –

In this project, I successfully built a SOC home lab environment using Splunk Enterprise, a Windows 10 virtual machine, and Splunk Universal Forwarder. The purpose of this project was to understand how security events are generated, collected, forwarded, and analyzed in a Security Information and Event Management (SIEM) platform.

During the project, I configured log forwarding from the Windows endpoint to Splunk and verified that security events were being successfully ingested. I then generated multiple security-related activities, including user account creation, user account deletion, privilege escalation, failed login attempts, successful logins, Remote Desktop access, PowerShell activity. For each activity, I verified the event generation in Windows Event Viewer and analyzed the corresponding logs in Splunk using SPL queries.

One of the most valuable parts of this project was troubleshooting and resolving log collection issues. This helped me understand how Windows Event Logs, and Splunk work together to provide visibility into endpoint activity. I also gained hands-on experience with event investigation and learned how different Event IDs can be used to identify user actions, authentication events, administrative changes, and process execution.

Through this project, I developed practical skills in log collection, event analysis, Windows security monitoring, PowerShell monitoring, and Splunk investigation workflows. The project gave me a better understanding of how SOC analysts monitor and investigate security events in real-world environments.

Overall, this project strengthened my knowledge of SIEM technologies and endpoint monitoring while providing practical experience with the day-to-day tasks performed by security analysts. It also established a strong foundation for future work in detection engineering, threat hunting, alert creation, and security operations.